Advanced Visualization

Molecular Data Science

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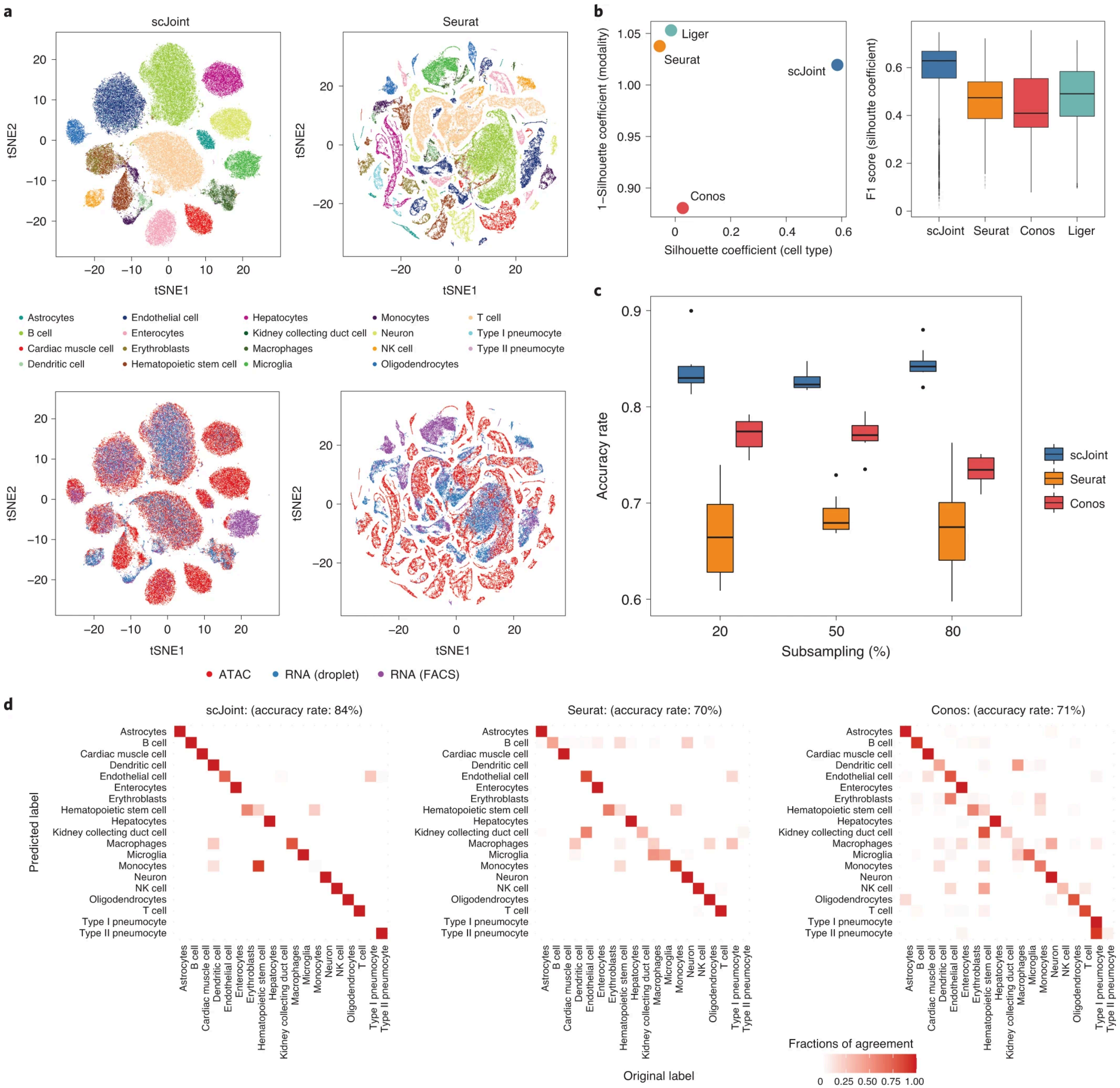
Overview

- The power of visualization
- What is a plot?
- Example: Differential equations and bacterial growth
- Example: Muscle cell differentiation
- Bar plots
- Line and point plots
- Visualization of distributions
- Sets: Venn diagrams, upset plots
- Multidimensional data
- ggplot
- Colors and file types

Visualization

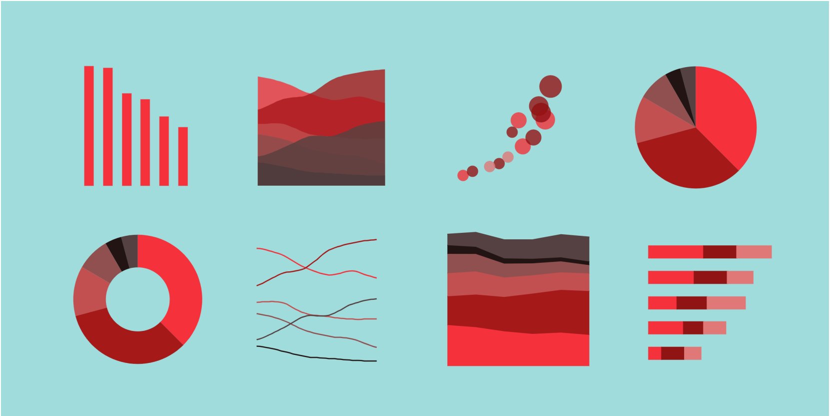
Data visualization is the representation of the data in a graphical format



from scJoint integrates atlas-scale single-cell RNA-seq and ATAC-seq data with transfer learning

The power of visualization

- *Explore*, comprehend and explain patterns and trends
- *Detect* patterns and anomalies in large chunks of data
- *Make sense* of large amounts of data efficiently and in time
- *Communicate* and share results and insights to a broad audience with ease



Main elements in a common plot

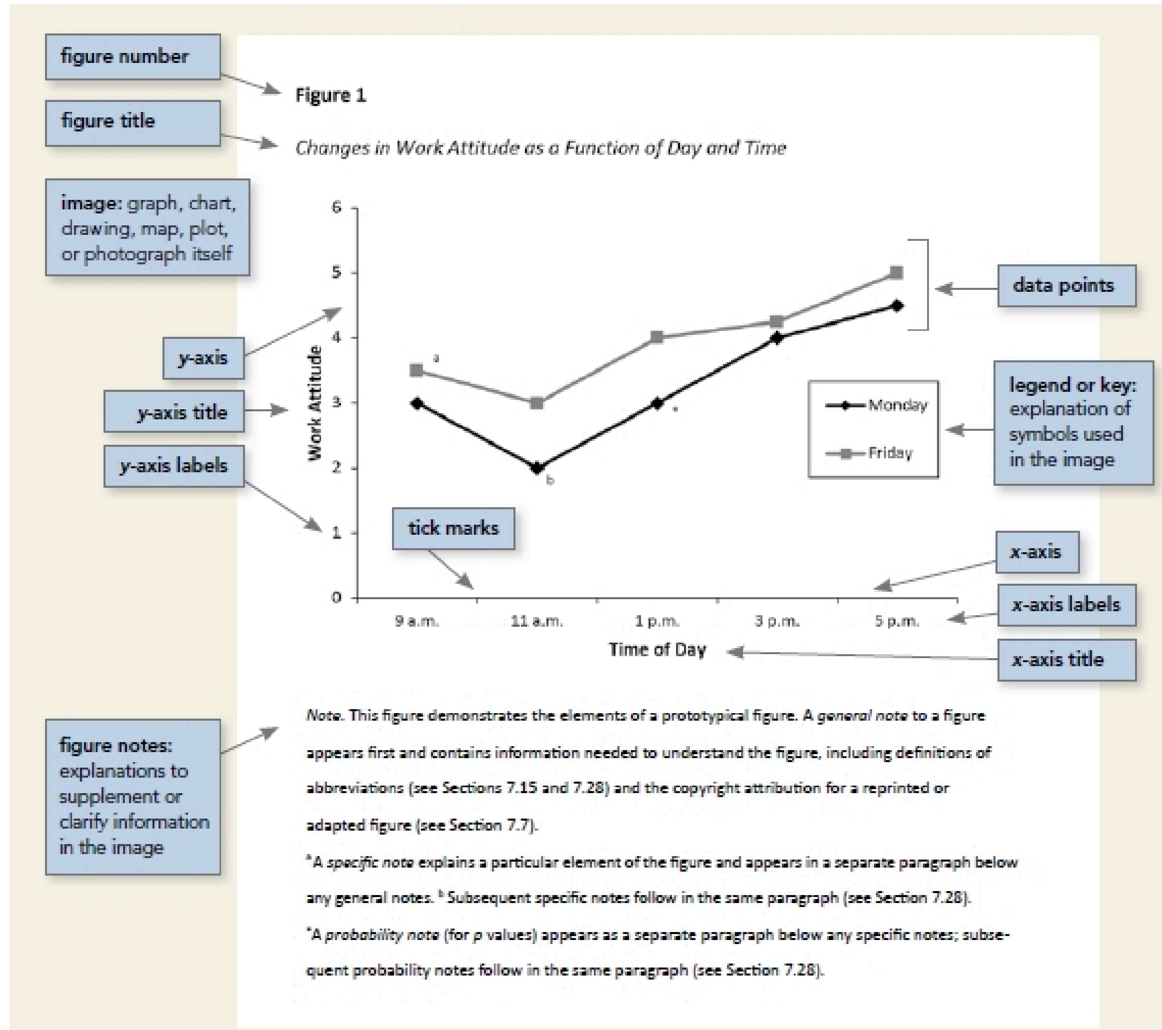
Title: What is the plot about?

Axes: What are the variables and their units?

Data points: What are the values and how are they distributed?

Legend: What do the different colors, shapes and sizes mean?

Ticks: Where are the values on the axes?



Bacterial growth

Can be described by a logistic growth with the underlying differential equation:

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$$

P is the number of bacteria, t is time, r is the growth rate and K is the carrying capacity.

How to solve it: link

Solution:

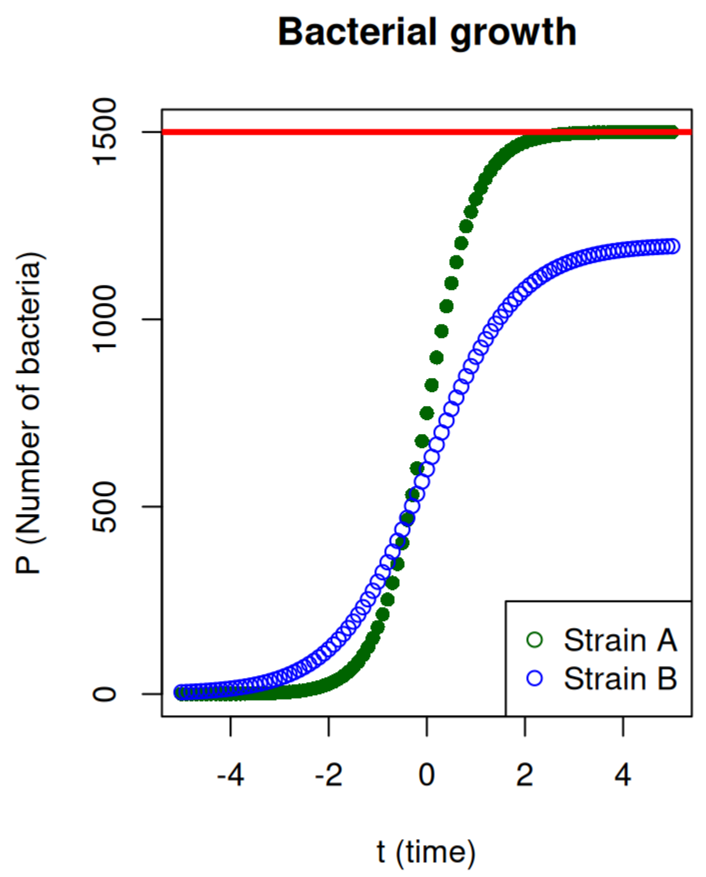
$$P(t) = \frac{K}{1 + \left(\frac{K - P_0}{P_0}\right) e^{-rt}}$$

Visualize some of the solutions

Application cases:

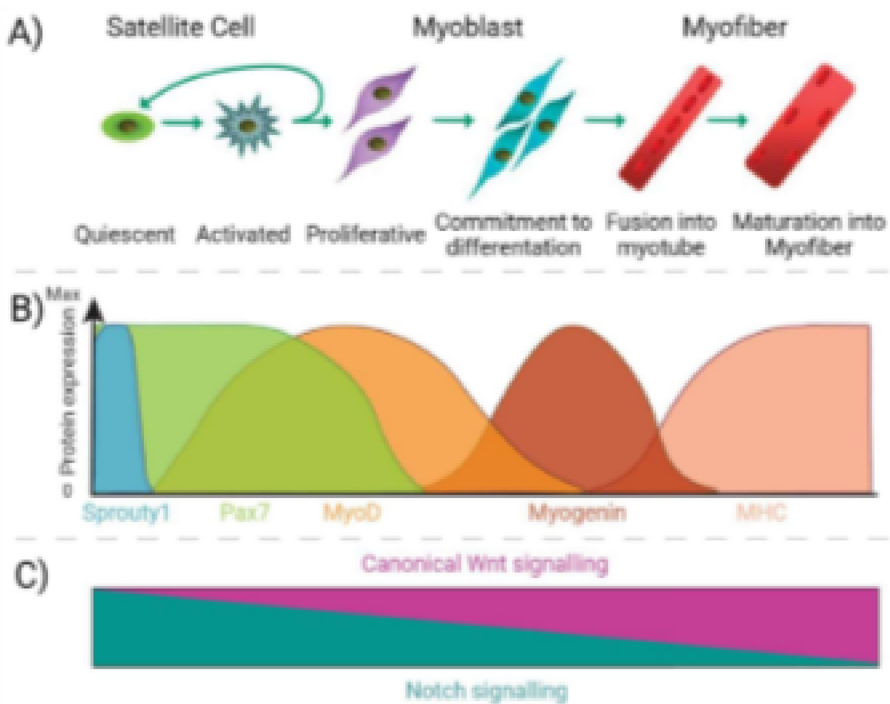
- Predict behavior of an experiment
- Compare your data to the logistic model
- Compare different bacterial growths
- Adjust experimental conditions to reach a certain growth behavior

▶ Code



Use case: muscle differentiation

We get protein abundances for:



a) almost 7000 different proteins

b) at 6 different differentiation stages (time points)

c) each measurement done three times (triplicates)

Data table

Download CSV

Search:

	X	Protein.Group	Protein.Name	Gene.Name	Alt.Gene.Name	Total.identified.peptides	Total.quantified.peptides	Day_ -1 a	
1	1	A0A0B4J2D5|P0DPI2	Glutamine amidotransferase-like class 1 domain-containing protein 3B, mitochondrial (Keio novel protein-I) (KNP-I) (Protein GT335) (Protein HES1)	GATD3B	GATD3A	12	12	-0.525117866699027	-0.
2	2	A0A0B4J2F2	Probable serine/threonine-protein kinase SIK1B (EC 2.7.11.1) (Salt-inducible kinase 1B)	SIK1B		6	5	0.147512580448094	0.
3	3	A0A0U1RRE5	Negative regulator of P-body association (P-body dissociating protein) (Protein NoBody)	NBDY		2	2	-0.187837654371373	-(
4	4	A0A0U1RRL7	Protein MMP24OS (MMP24 opposite strand)	MMP24OS		2	2	-0.139907833704359	-0.0
5	5	A0A1B0GTQ4	Protein myomixer (Microprotein inducer of fusion) (Protein minion) (hMINION)	MYMX		3	3	-1.62468808555076	-:
6	6	A0AVT1	Ubiquitin-like modifier-activating enzyme 6 (Ubiquitin-activating enzyme 6) (EC 6.2.1.45) (Monocyte protein 4) (MOP-4) (Ubiquitin-activating enzyme E1-like protein 2) (E1-L2)	UBA6		33	26	0.133705210300527	0.0

Showing 1 to 6 of 6,863 entries

Previous 1 2 3 4 5 … 1,144 Next

What do we want to visualize?

Quality control:

- Do we have similar numbers of proteins per sample?
- How consistent are the different measurements?

Biological insights:

- How do single proteins in the table change during cell differentiation?
- How much does the number of quantified proteins change within same sample type / experimental condition?
- How does the abundance of the proteasome complex change during differentiation?
- How many proteasome proteins increase/decrease in their abundance *and* how much?
- How consistently are up- and down-regulated proteins showing higher changes in their abundance?

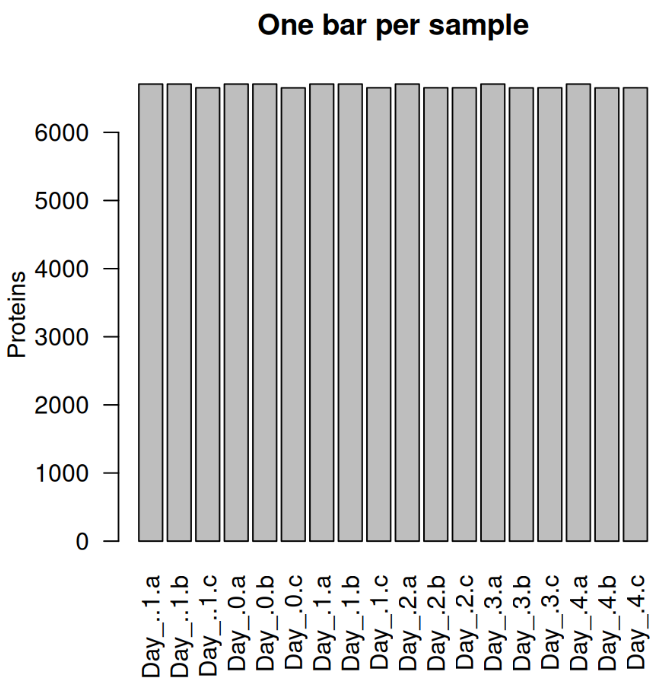
Bar plots

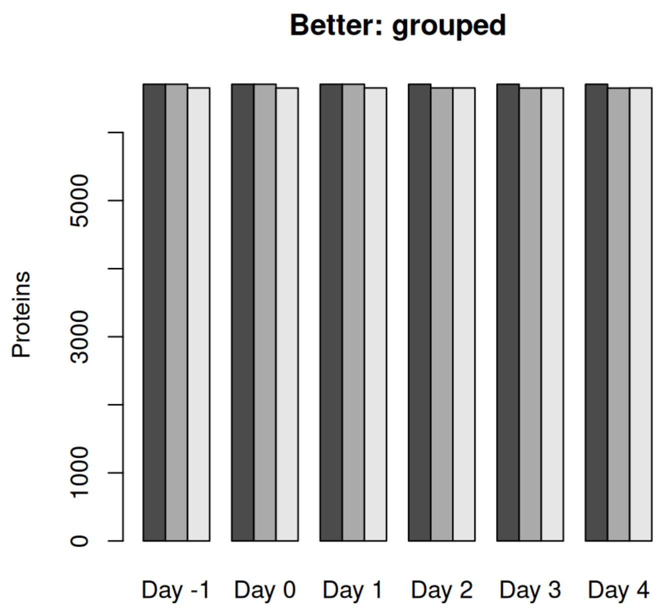
Have we found similar numbers of proteins per sample?

or

How consistent are the different measurements?

▶ Code





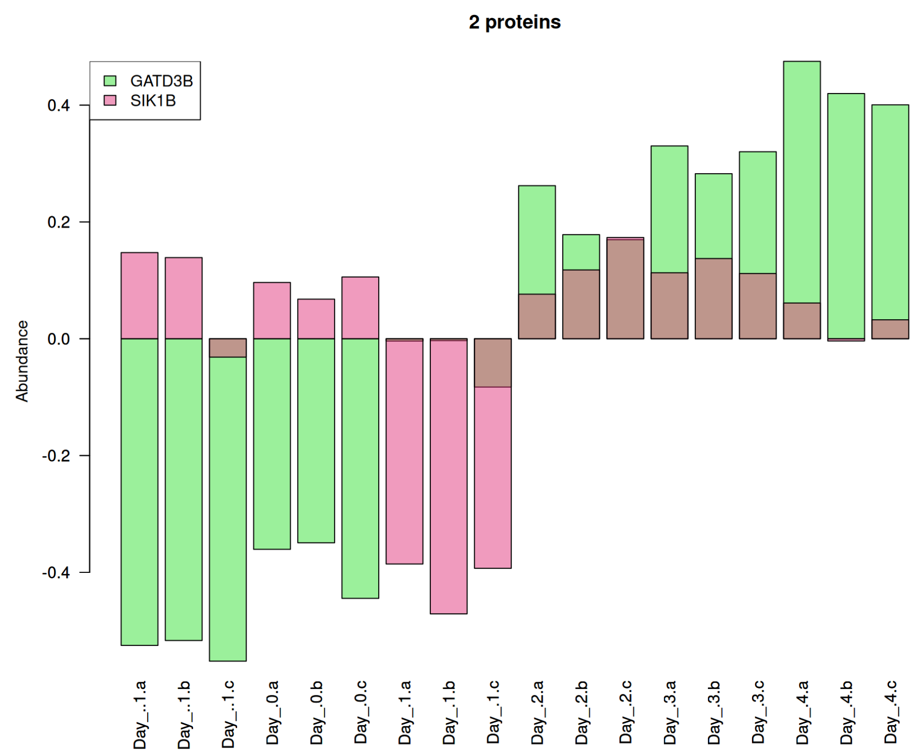
Bar plots

What do we display?
Single protein abundances compared across different types and categories. Can be subcategories, too.

Different layouts
Colors, stacked, grouped, beside. You can use transparency to overlay them.

For example: How do the first two proteins in the table change during cell differentiation?

▶ Code

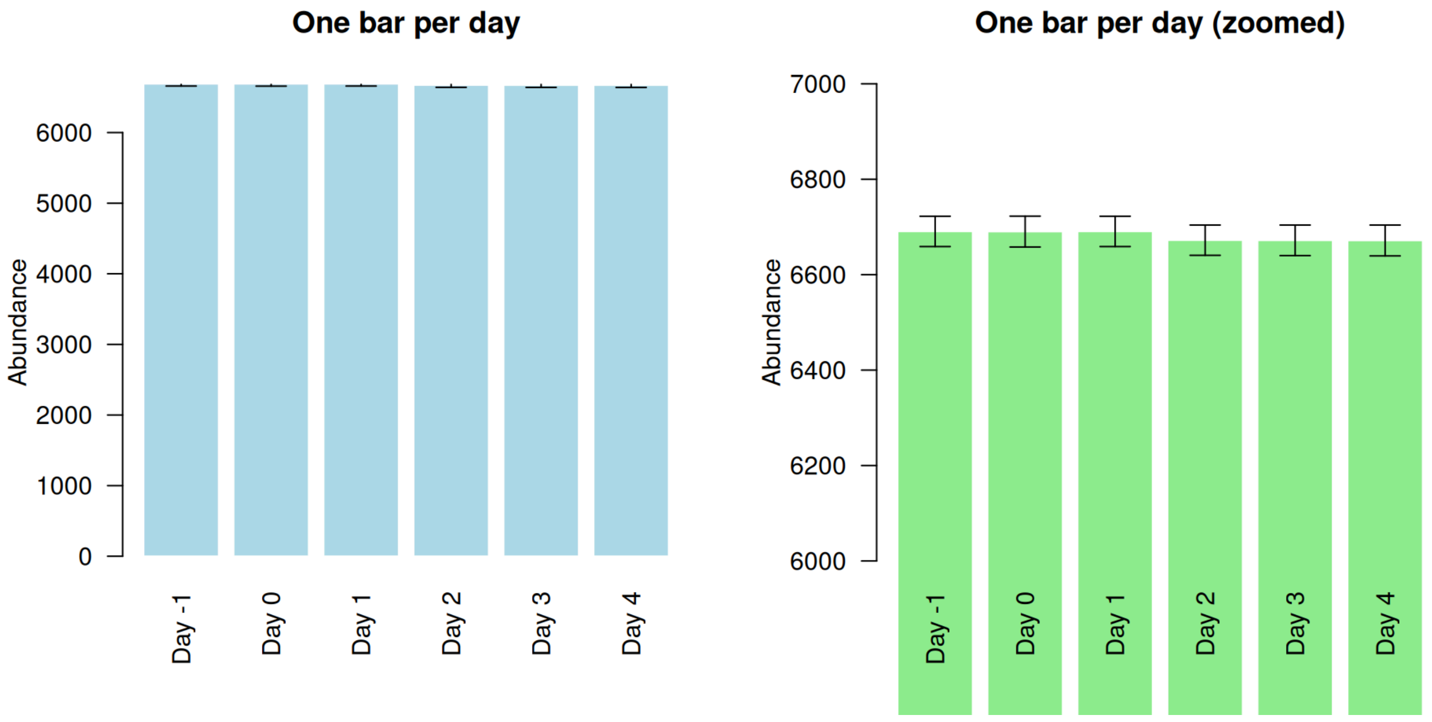


Bar plot with error bars

Use standard deviation to display variation within samples

How much does the number of quantified proteins change within same sample type / experimental condition?

▶ Code

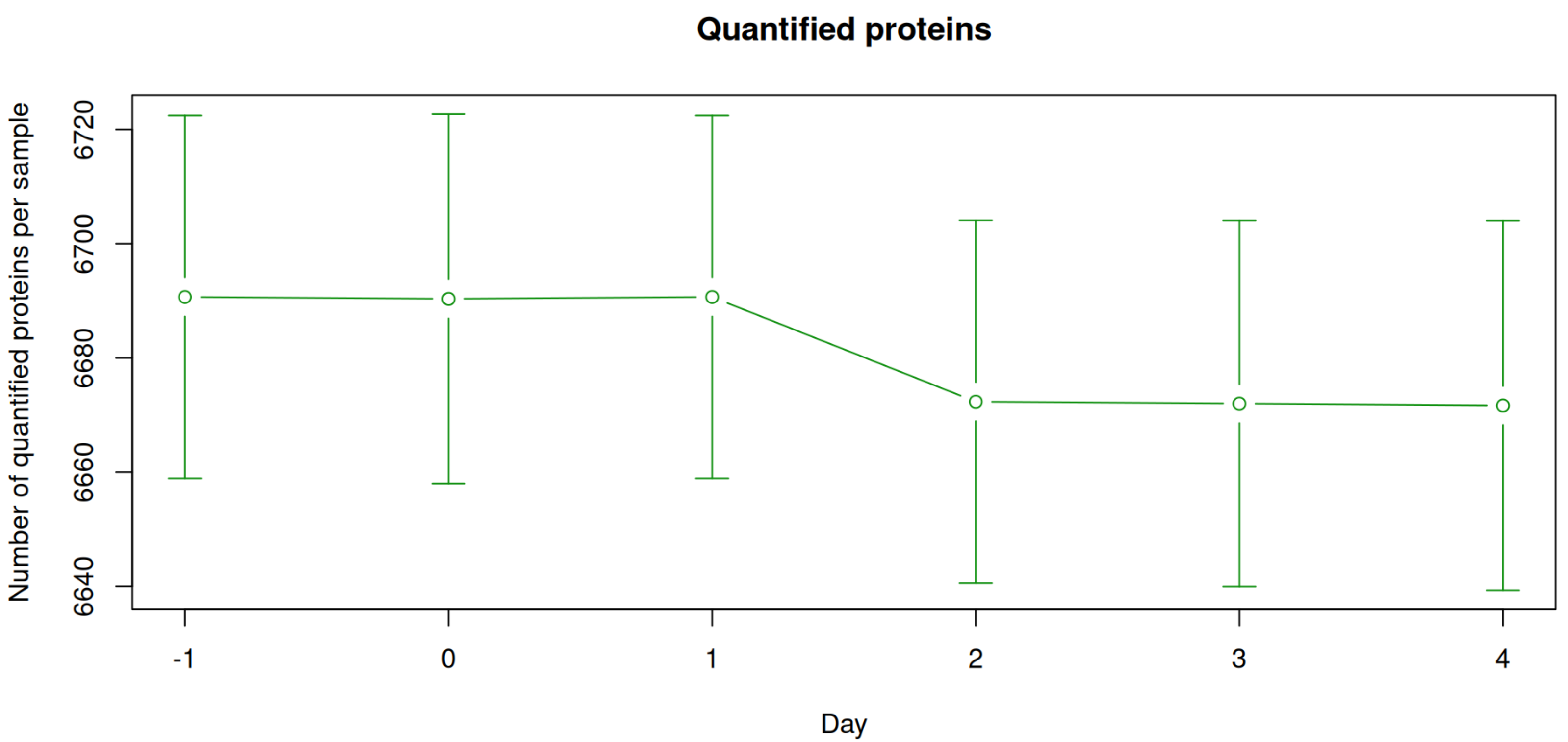


Line plots

Same data but more emphasis on error

Note the different scale!

▶ Code



Line plots

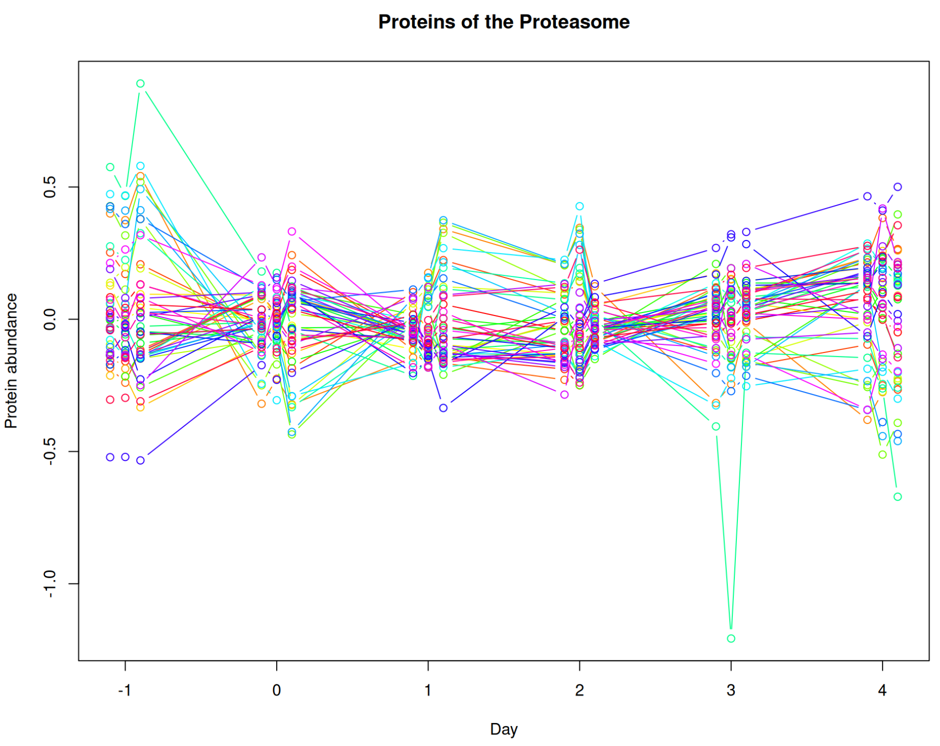
What do we display?
Lines or points given by *x* **and** *y* coordinates

Different layouts: Colors, lines, lines with points, different line and point types

Let’s look at the proteins of the Proteasome complex

How does the abundance of the proteasome complex change during differentiation?

▶ Code

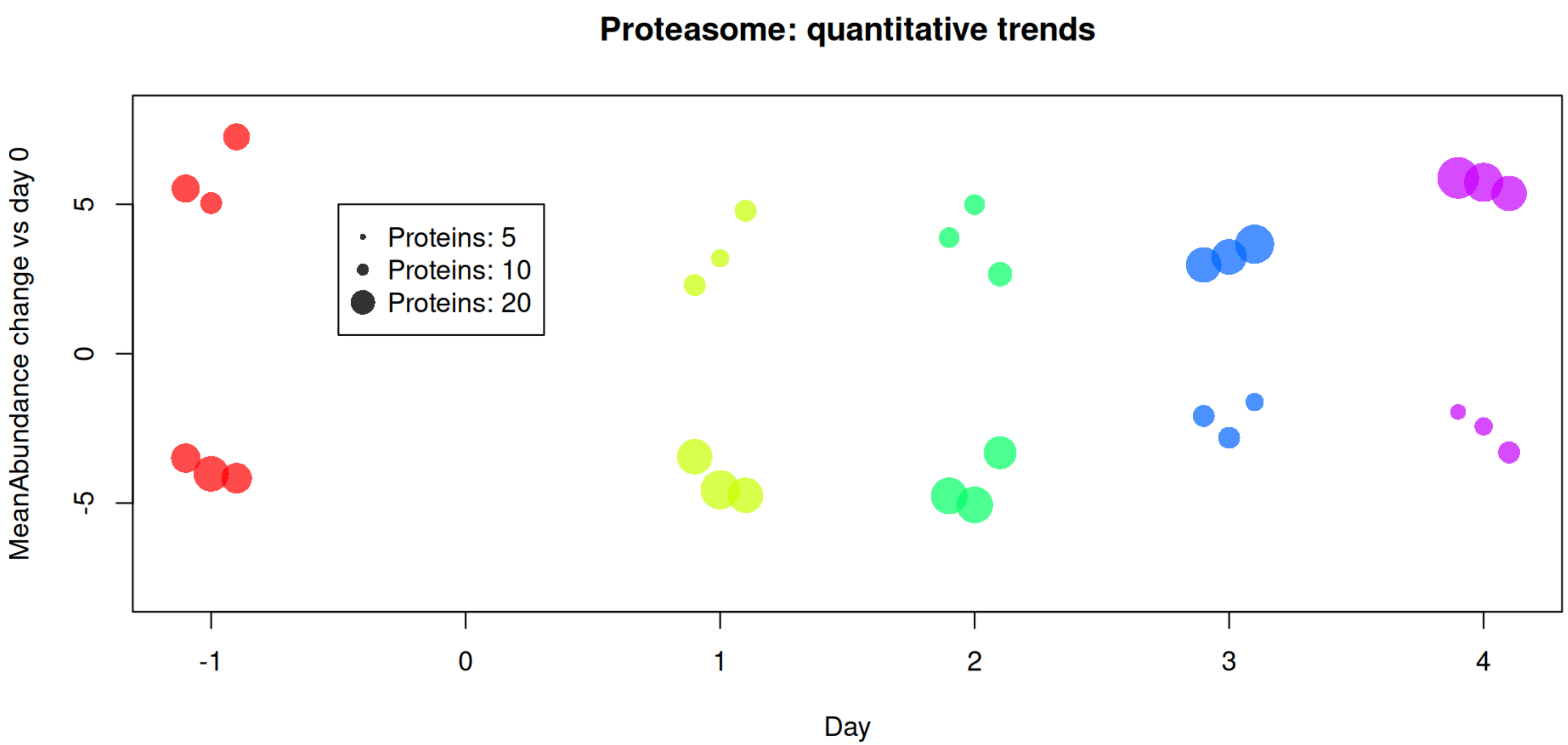


Bubble plot

Let's dig a bit deeper by comparing versus day zero.

<u>How many proteasome proteins increase/decrease in their abundance *and* how much?</u>

▶ Code



Volcano plot

Frequently used to summarize output from statistical tests like t-tests

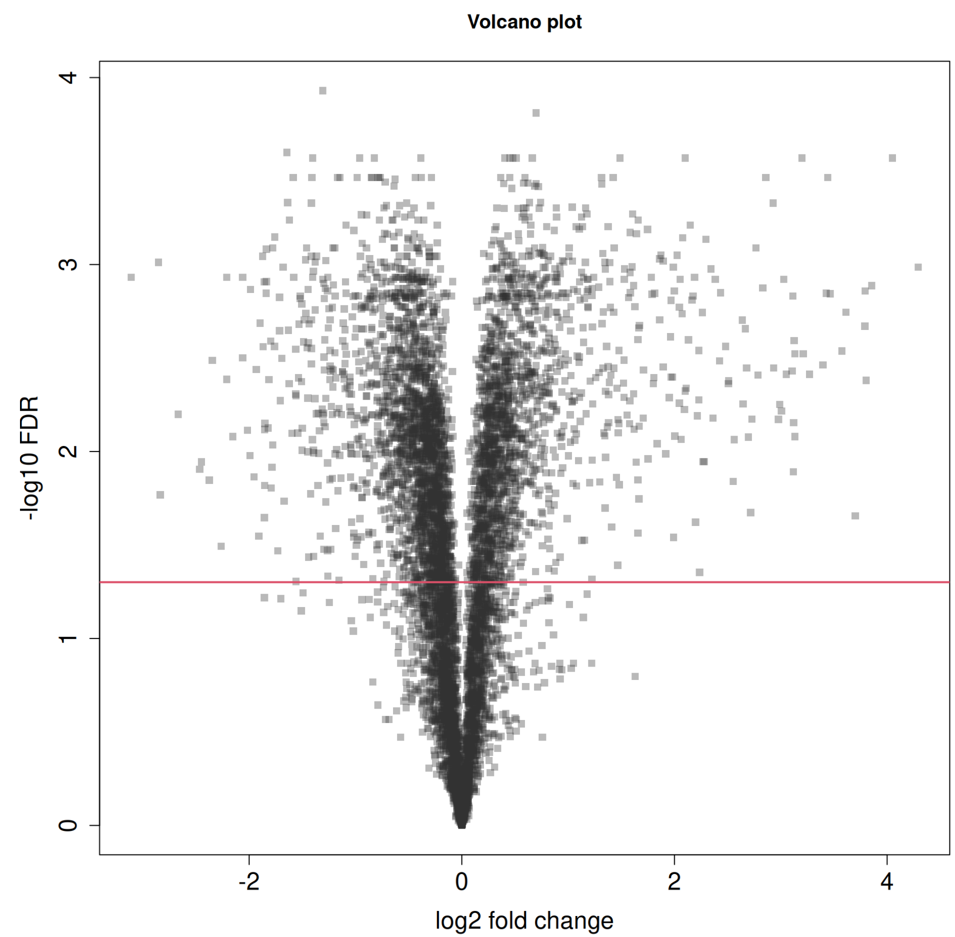
Scenario: We run one statistical test per protein $\longrightarrow$ 6863 p-values

X-axis is then the logarithmic fold-change

Y-axis denotes the significance for the differential regulation[1]

<u>How consistently are up- and down-regulated proteins showing higher changes in their abundance?</u>

▶ Code



Distributions

You already had the histogram, but there is more:

- Histogram(s)
- Density plots
- Boxplots
- Violin plots
- 2D density plots / histograms

Example: The 3 samples of the first measurement (day -1)

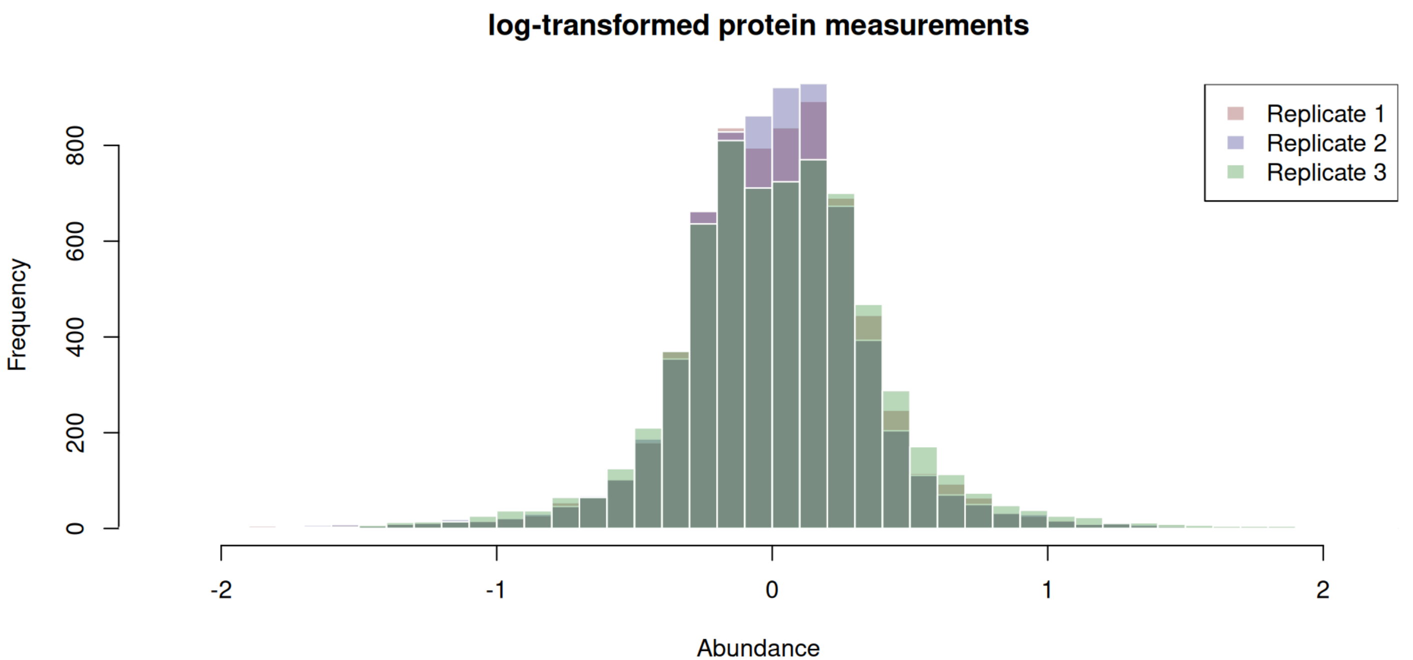
Histograms

Again, transparency helps overlaying distributions

How are the abundance values distributed?

Anything pointing to experimental issues?

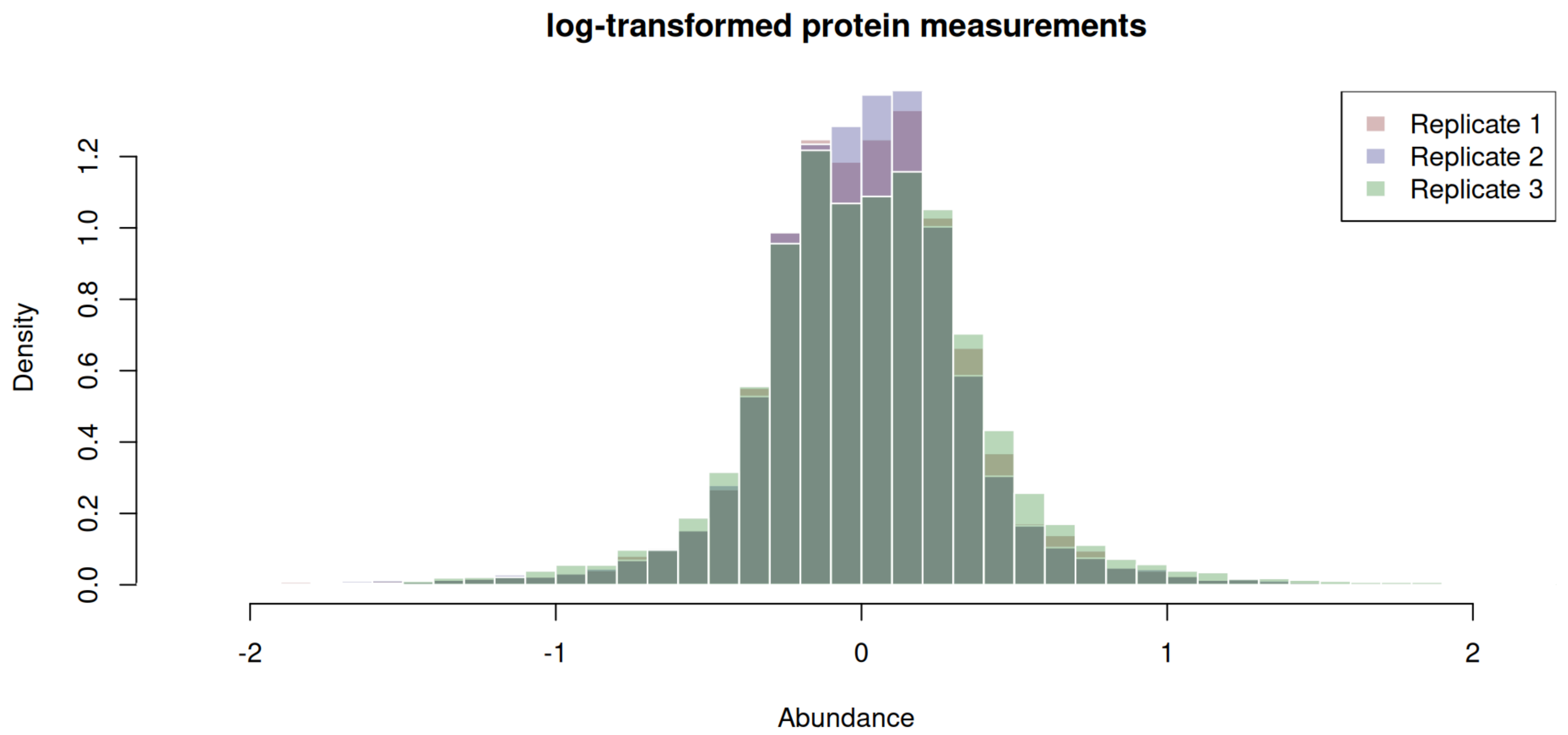
▶ Code



Histograms

Now probability and not frequency

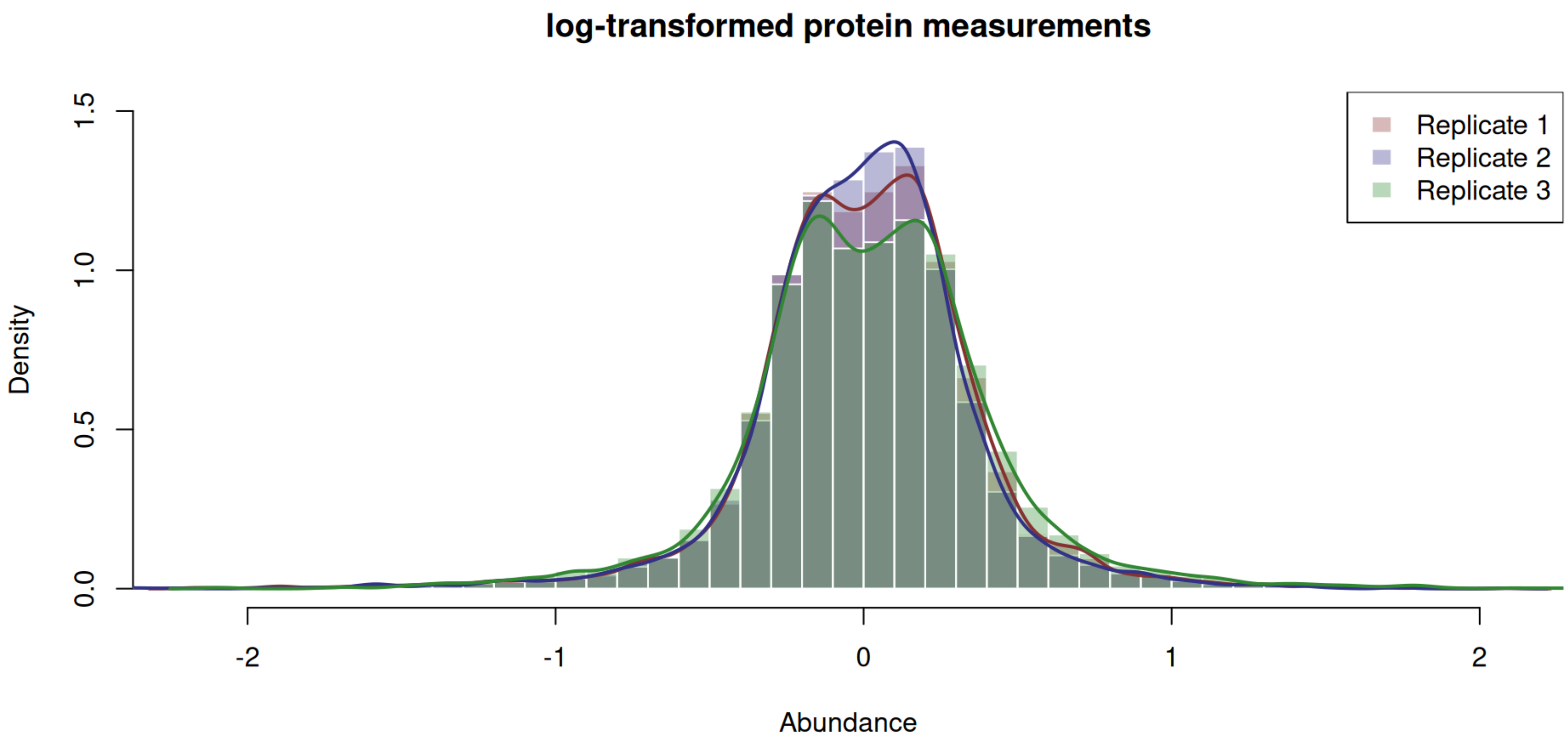
▶ Code



Histograms

Adding density plots

▶ Code

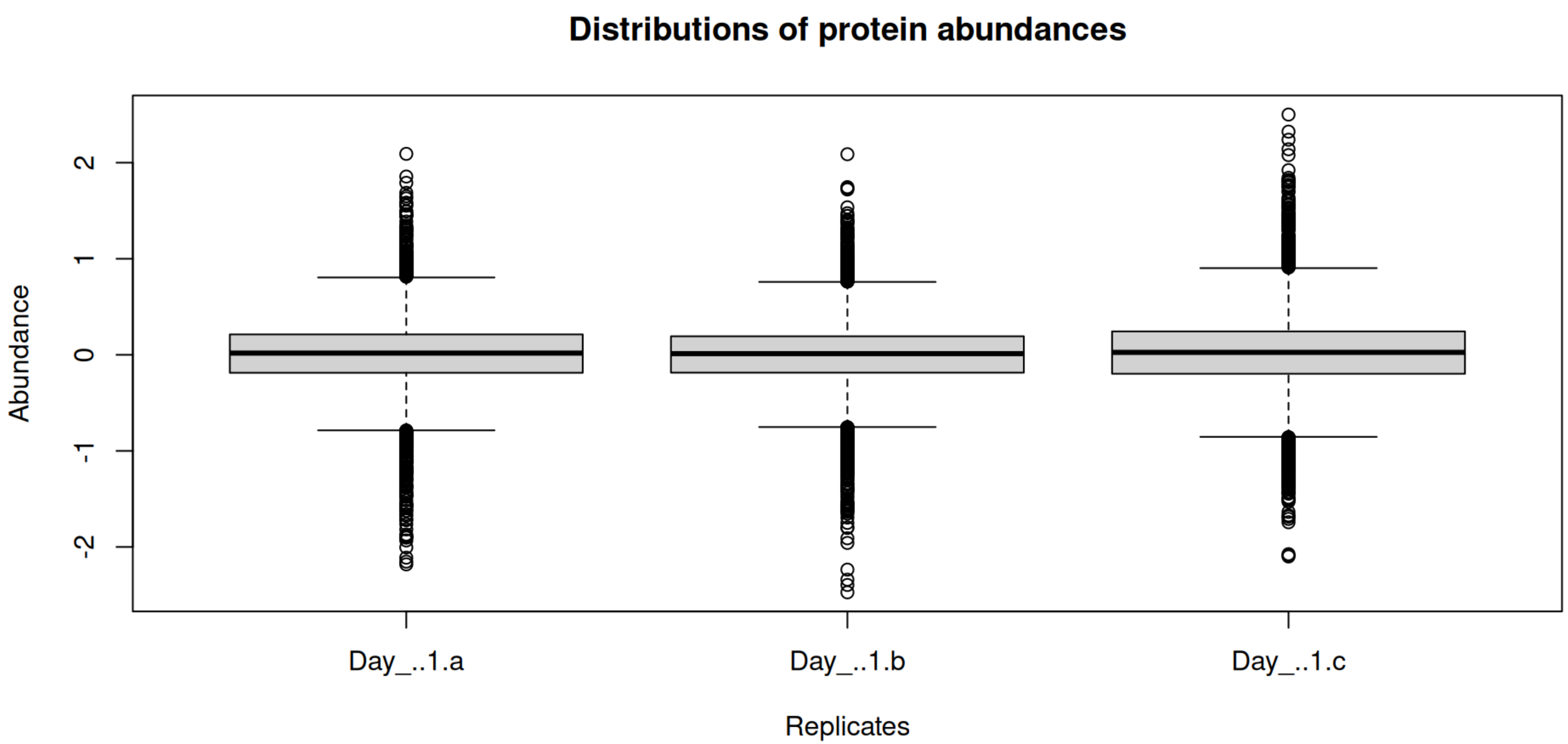


Boxplots

Better for comparing multiple distributions

Are there any unexpected changes in the general properties of the distributions?

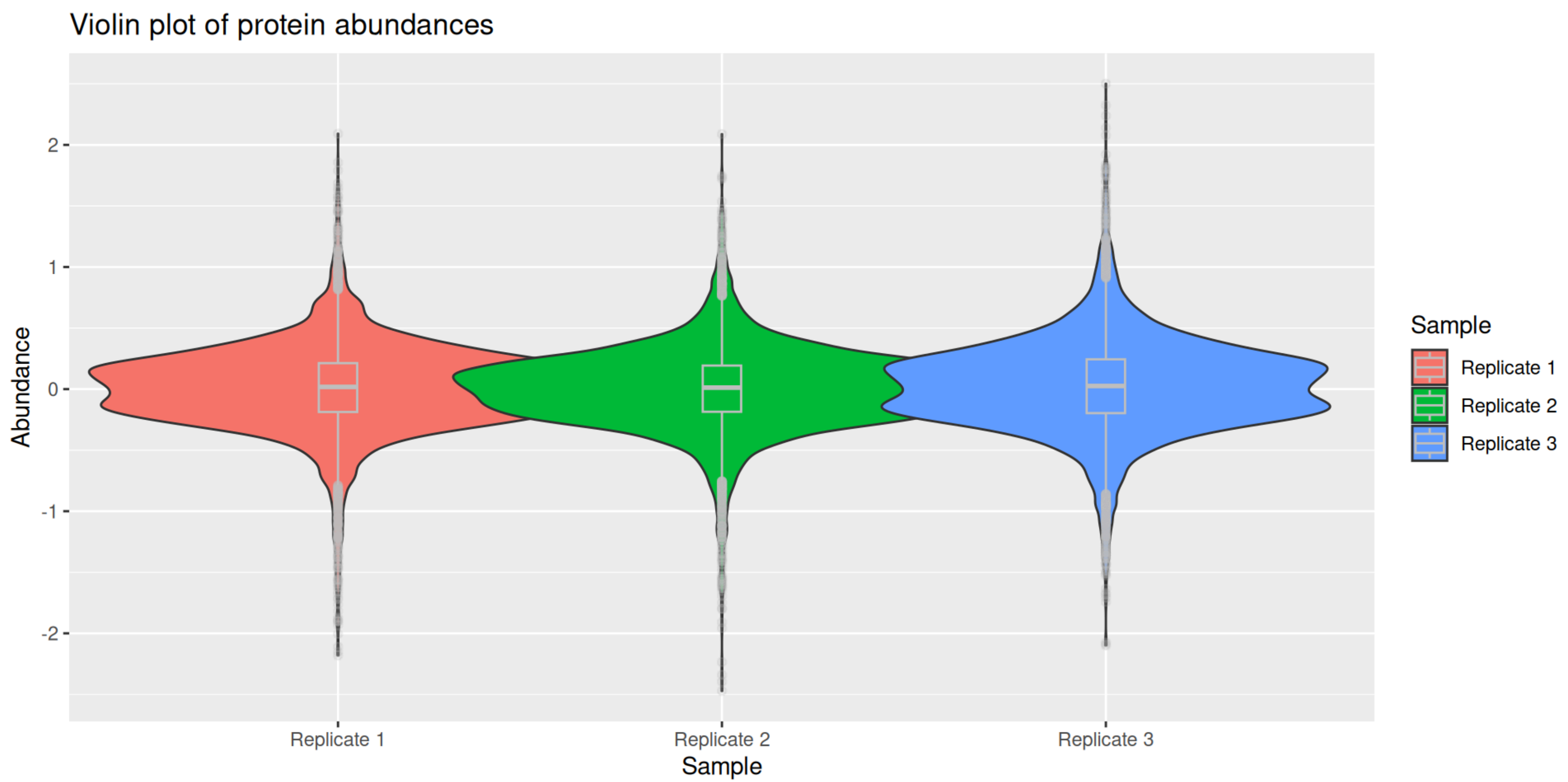
▶ Code



Violin plots

Elegant way to combine density plots with boxplots, each telling part of a story

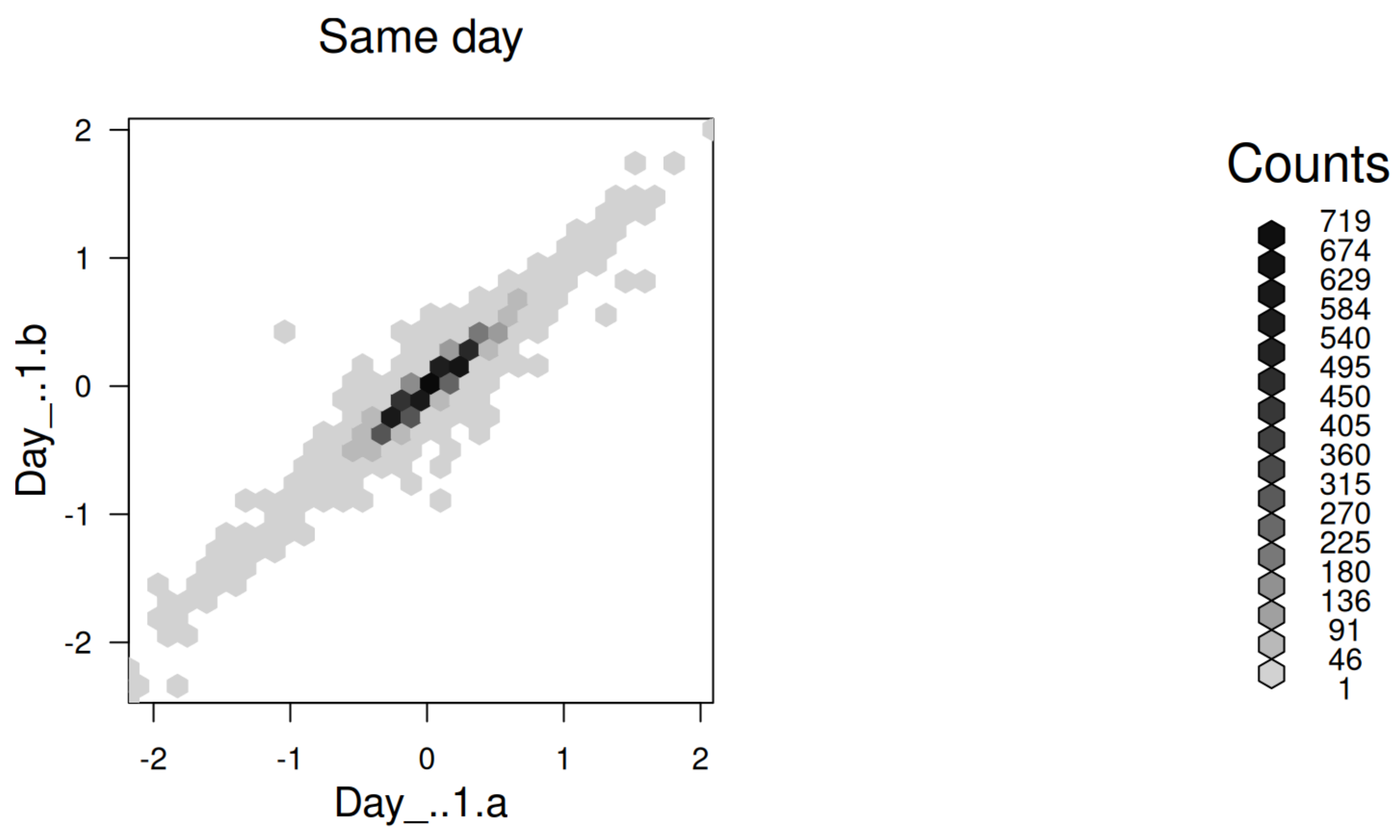
▶ Code



2-dimensional histograms

Are proteins with higher abundance also higher in the other replicate?

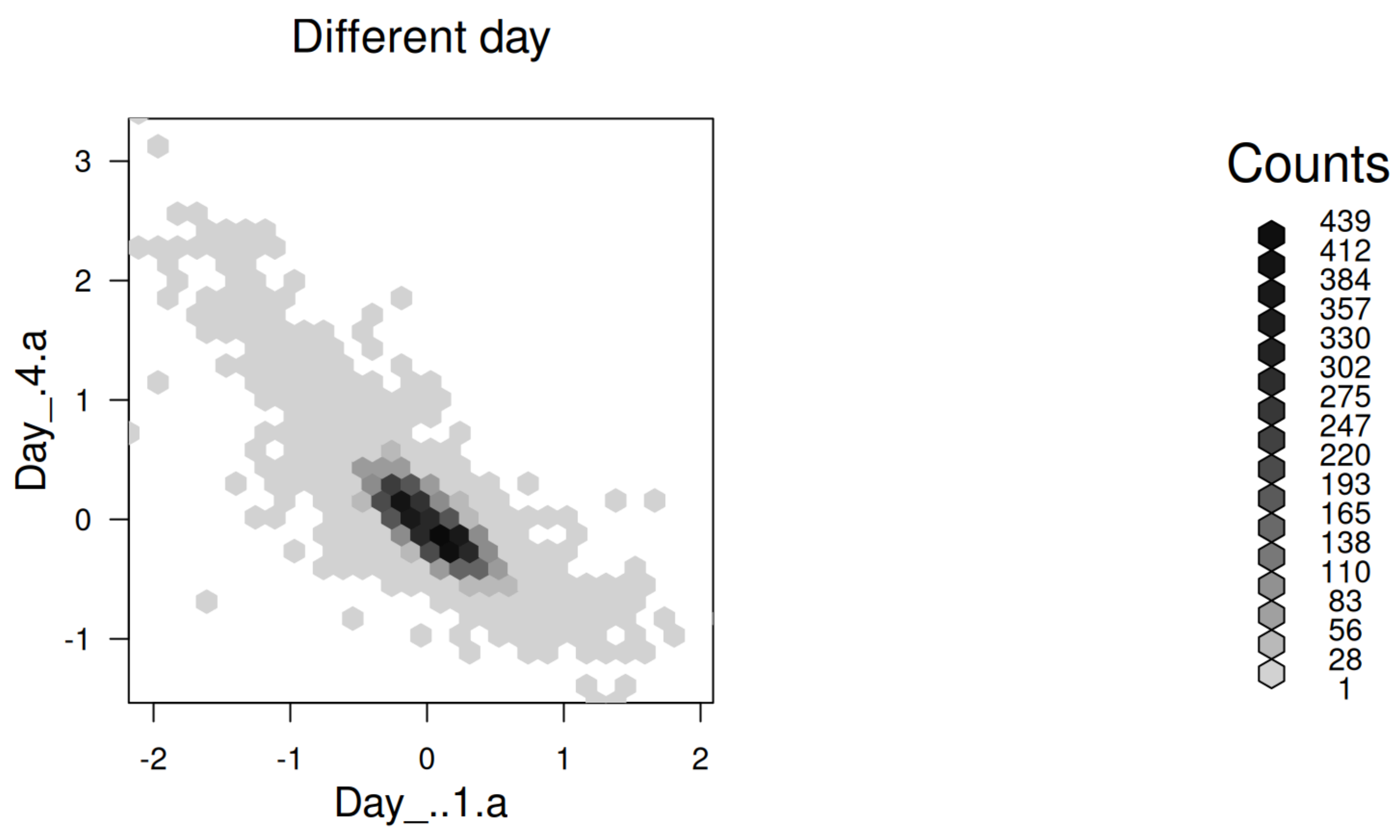
▶ Code



2-dimensional histograms

How much changes in protein abundance do we obtain when comparing the extremes?

▶ Code

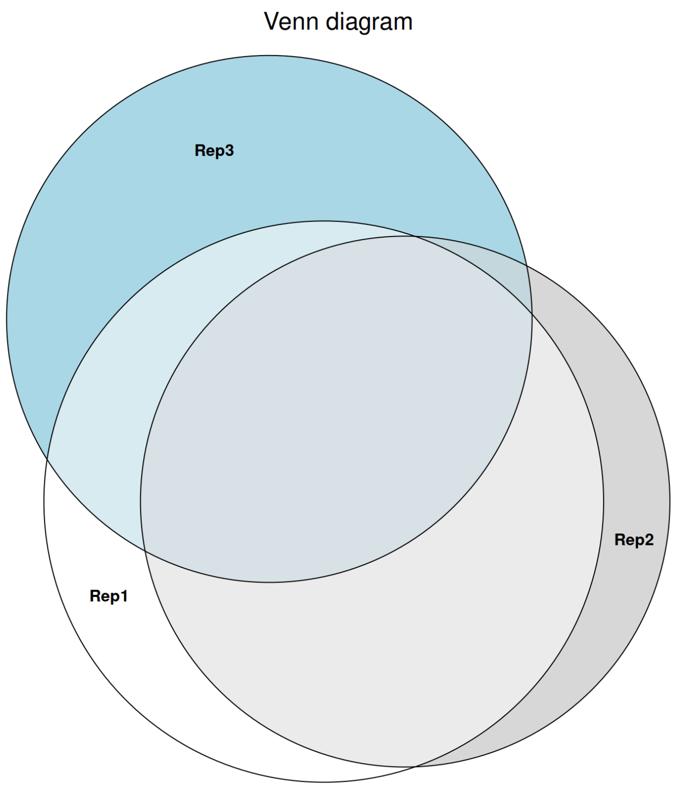


Sets

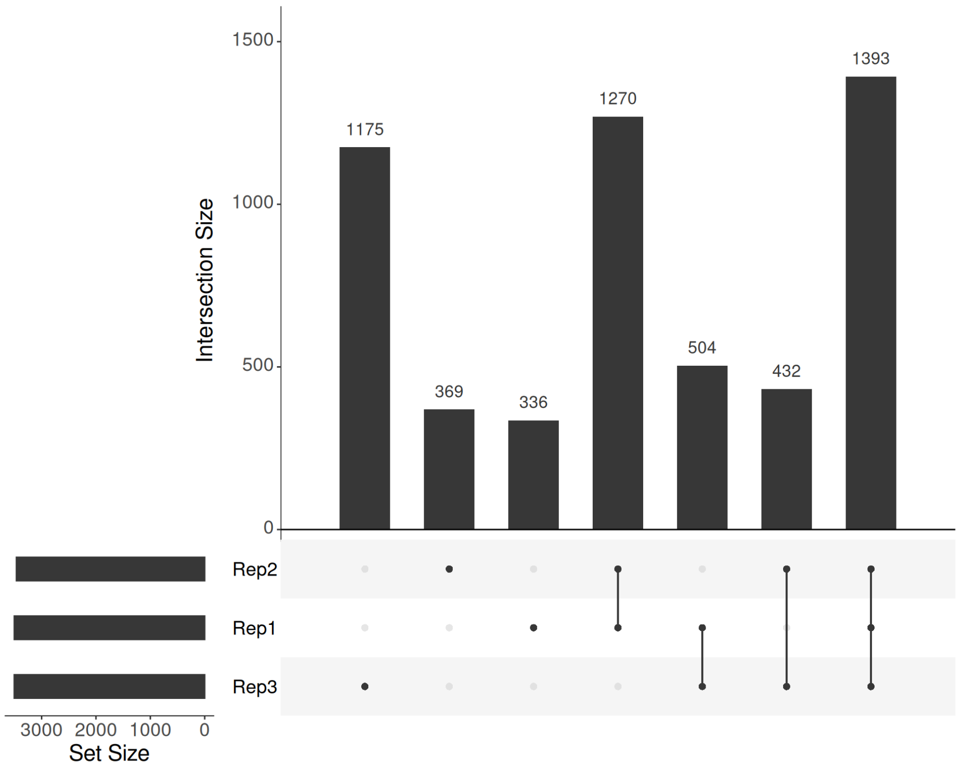
Venn diagrams are frequently used to compare 2-4 sets of numbers. You can give the numbers or the percentages of the intersection. Upset plot are a nice alternative.

How many proteins have abundance larger than zero when comparing the first replicates of the first 3 days?

▶ Code


Venn diagram
Rep3
Rep1
Rep2

▶ Code


Intersection Size
1500
1000
500
0
1175
369
336
1270
504
432
1393
Rep2
Rep1
Rep3
3000
2000
1000
0
Set Size

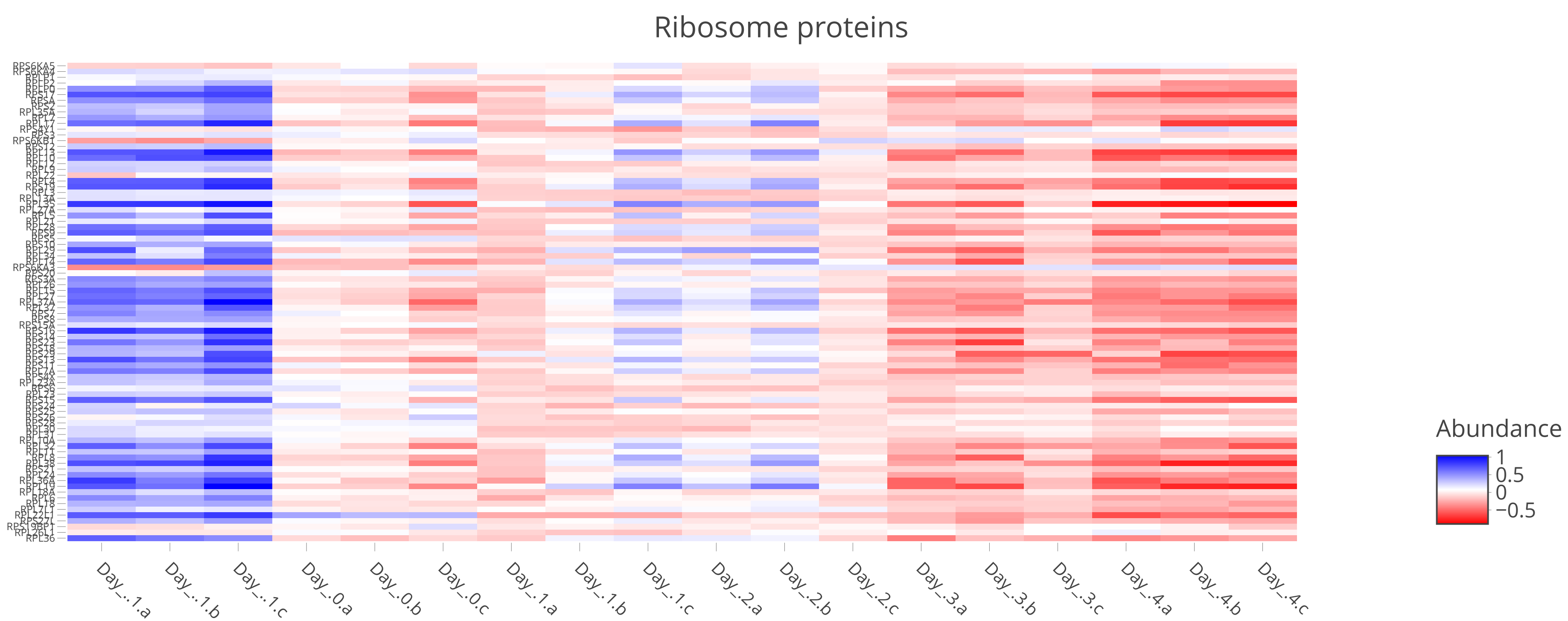
Visualization of multivariate data

We can use colors more efficiently. Color now corresponds to abundance.

Ribosome proteins decrease in their abundance at later stages of differentiation.

What could be the biological reason for this behavior?

▶ Code

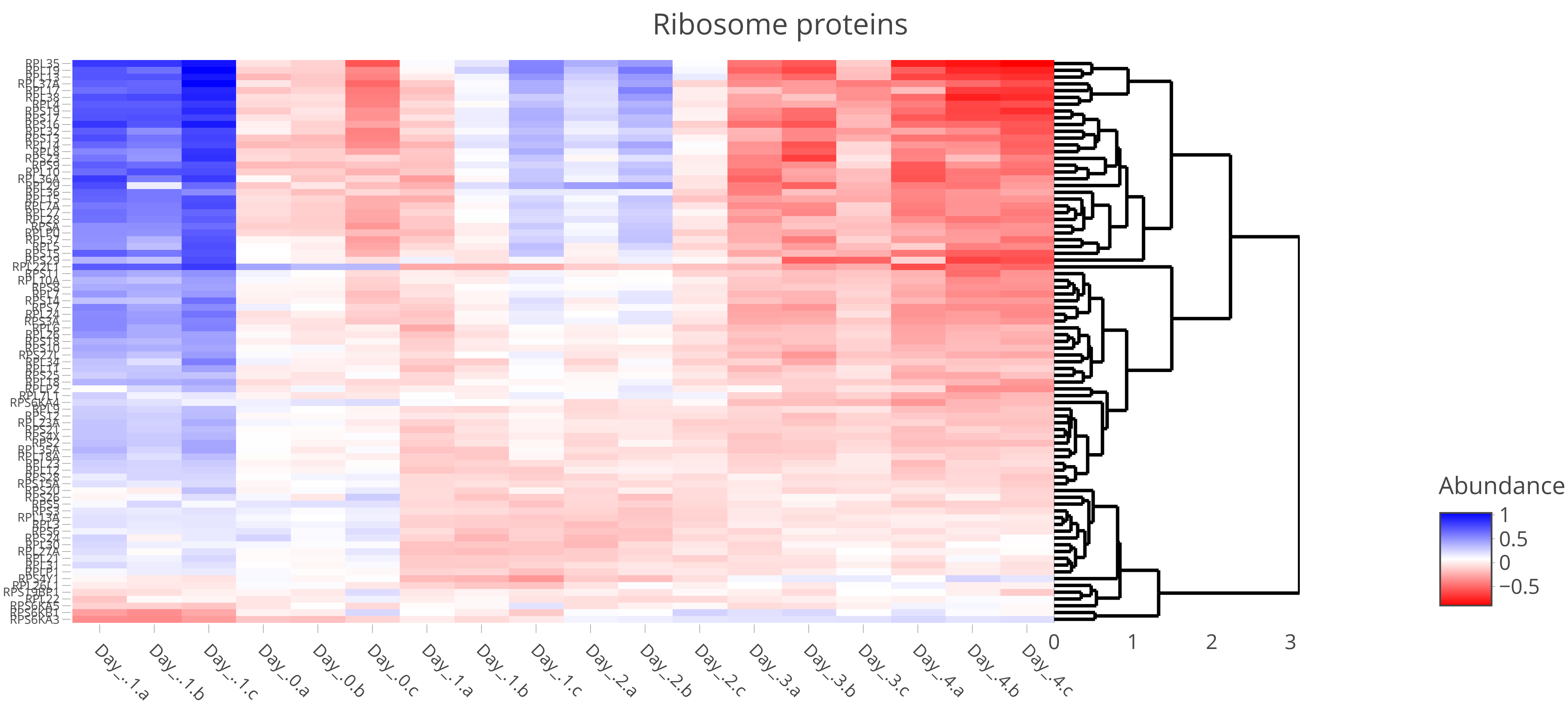


Visualization of multivariate data

Better: order proteins according to similarity of their abundance changes

Now we can see that there is a strongly co-regulated subgroup.

▶ Code



Different "philosophies"

You can create even the same type of figures multiple ways. See the R Graph Gallery for example. Here *ggplot* is one of the major players.

However, you will need to arrange the data differently:

Wide Format

Team	Points	Assists	Rebounds
A	88	12	22
B	91	17	28
C	99	24	30
D	94	28	31

Long Format

Team	Variable	Value
A	Points	88
A	Assists	12
A	Rebounds	22
B	Points	91
B	Assists	17
B	Rebounds	28
C	Points	99
C	Assists	24
C	Rebounds	30
D	Points	94
D	Assists	28
D	Rebounds	31

Find more here

File formats and rendering

There is:

- Bitmap formats like gif, png and jpeg
    - Resolution very important
    - All built from the filling in pixels
    - Good for photos and pictures that contain many details
- Vector graphics like svg, pdf and wmf
    - Infinite resolution
    - Different elements (e.g. arrows) added to empty sheet
    - Preferred in scientific plotting
    - Can become large when for example plotting 1 million data points

Zoom into the figures of this publication